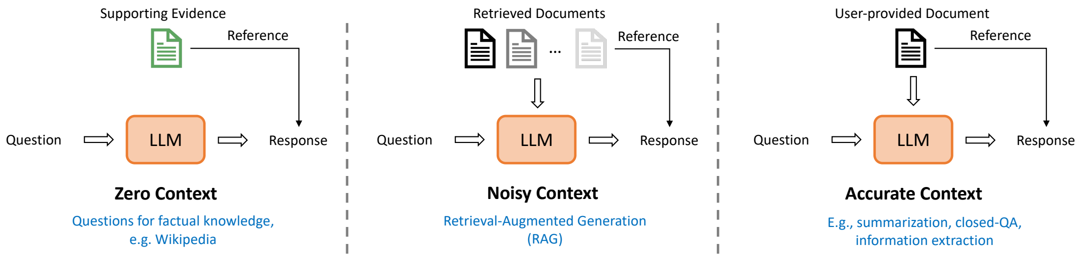

Figure 3: Illustration of three settings of context, tasks and references. Zero Context is about seeking factual knowledge from the internal memory of the LLMs. Noisy Context has context information retrieved from a knowledge source, which is a RAG use case. Accurate Context has context provided in the input prompt. For Noisy and Accurate Context, we take the input context as the reference.

serve as prompts to perform the closed-book question answering task, and the annotated long answer as the reference for a fair evaluation of LLMs and comparison between different approaches. These references are paragraphs from Wikipedia articles, which are widely used for pre-training LLMs (Zhao et al., 2023), making them a suitable proxy for the internal knowledge of these LLMs.

Noisy Context (NC) In this setup, the LLM receives additional context retrieved from some external knowledge source, which may contain noisy or irrelevant information. NC is also known as RAG, a crucial use case frequently encountered in real-world applications. We utilize questions sourced from MS MARCO (Nguyen et al., 2016). Each question in this dataset is accompanied by a list of documents retrieved from the internet, serving as the input context.

Accurate Context (AC) This setting is similar to NC but the reference is typically noise-free. Examples include text summarization, closed-QA and information extraction tasks. We employ a subset from the databricks-dolly-15k (Conover et al., 2023) instruction tuning dataset which covers the 3 tasks mentioned before.

For both Noisy and Accurate Context settings, the prompt is the reference followed by the query, whereas for Zero Context, the prompt is just the query (Figure 3). The benchmark contains 300 examples in total, 100 for each setting, and we summarize it in Table 6. The details of the benchmark curation process are described in Appendix A.

3.2 Claim-Triplets and Definition of Hallucination

Informally, claims are the units for the checking. This work explores the approach of representing claims with knowledge triplets. This concept is inspired by the field of knowledge graph studies,

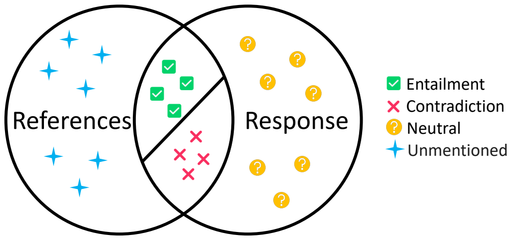

Figure 4: Definition of fine-grained hallucinations in an LLM-generated response compared with references. The intersections of the response and the references are the claims either supported (Entailment) or refuted (Contradiction) by the references. The remaining parts in the response are claims not verifiable by the references (Neutral). The other parts of the references are the content not mentioned in the response.

where triplets are employed to encapsulate factual knowledge units. Knowledge triplets adopt a `(head_entity, relation, tail_entity)` structure to capture fine-grained information within the response. We call the triplet-format claims as *claim-triplets*, examples of which are shown in Figure 2.

Subsequently, the claim-triplets are compared with a reference to determine the type of hallucinations, as illustrated in Figure 4. If a claim-triplet can be directly inferred from the reference, we classify it as **Entailment**. Conversely, if it contradicts the information in the reference, it is labeled as **Contradiction**. However, in cases where the reference is insufficient to verify the claim-triplets, we classify it as **Neutral**. In this study, we focus on verifying hallucinations in the response and do not consider unmentioned aspects in the reference, which may also be important for certain tasks.

3.3 Human Evaluation

We performed a human evaluation of responses generated by seven LLMs on this benchmark dataset, including GPT-4, GPT-3.5-Turbo (OpenAI, 2022), InstructGPT (text-davinci-001) (Ouyang et al., 2022), Claude 2, Llama 2 70B Chat, Falcon

40B Instruct (Almazrouei et al., 2023) and Alpaca 7B (Taori et al., 2023). The process involves three steps: gathering responses, extracting claim-triplets with an extractor, and asking human annotators to evaluate these claim-triplets. We annotated a total of 11k claim-triplets for 2.1k responses. 23% of the claim-triplets were double annotated, with 95.0% Inter-Annotator Agreement. See Appendix A.3 for the details of the annotation process.

A significant finding from the human evaluation is the crucial role of contextual information for factual responses. On average, the rate of Contradiction decreased from 25% in the absence of contextual cues (ZC) to 13% with NC, and further reduced to 6% with AC, which reflects the necessity of distinguishing the three context settings for separate study (cf. Appendix A.4).

4 RefChecker Framework

As illustrated in Figure 2, the RefChecker framework is designed as a 2-stage pipeline: an Extractor E decomposes the LLM response into a set of triplets, with each of them verified by the Checker C. The categorization of the triplets can be optionally aggregated according to specified rules. We explain them in the subsequent subsections.

4.1 Extractor

Our checking framework hinges on a key assumption: the decomposition of the original text into triplets facilitates finer-grained detection and more accurate evaluation. The extraction of these triplets plays a pivotal role in achieving this objective. We apply LLMs to extract knowledge triplets from the given text. We began with GPT-4 and Claude 2 and, for both cost and efficiency concern, Mixtral 8x7B and Mistral. More specifically, we performed knowledge distillation to train a 7B Mistral-based extractor with Mixtral 8x7B as the teacher. We conducted supervised fine-tuning on 10k responses. Evaluation in Sec. 6.1 shows competitive extraction quality of the open-source extractor. Please refer to Appendix B.1 for prompts used for extraction and details on extractor training.

4.2 Checker

We experimented with two families of checkers, the first is off-the-shelf LLMs including GPT-4 and Claude 2 (see Appendix B.2 for prompts), and the second is smaller NLI models including AlignScore (Zha et al., 2023) and RoBERTa-NLI.[2] Long references in AC/NC setting are split to fit into small context windows of these small models (e.g. 200 tokens), and the results are aggregated later.

Mistral 7B (Jiang et al., 2023), unlike GPT-4/Claude 2, performs poorly as a zero-shot checker, its performance improves with demonstrations but is still not satisfactory. However, the fact that the model weight is open gives us the opportunity to improve it by fine-tuning with NLI data. There are many options we have experimented: 1) fine-tune by adding small amount of new parameters using LoRA (LoRA-sft) (Hu et al., 2021), 2) attach a shallow classifier, eg. SVM, 2-layer MLP, KNN after NCA projection (Goldberger et al., 2004), on top of the internal states of the model. We call such checker RepC (for Representation-based Classifier). Such states can be selected from one layer (layer selection, LS) or an ensemble of all layers (layer ensemble, LE). As we will report in Sec. 6.2, RepC checkers are competitive in general.

4.3 Aggregation

Triplet results can be aggregated to obtain the ratio of each category, therefore gives an overall measure of hallucination distribution in a response. To derive the performance of a particular LLM, we take a macro average on Entailment/Neutral/Contradiction ratios of all responses. If a scalar is preferred, we can assign certain numeric values to the catogories, for instance $-1, 0, 1$ for contradictory, neutral and entail, respectively.

The aggregation can be customized and this is one of the benefits of the fine-grained hallucination design in RefChecker. For instance, to compare against other response-level approaches (cf. Sec. 5.1), we adopt a rule where the response is flagged as contradictory if any one of the claim triplet is contradictory.

5 Experiments

The major difference between RefChecker and other related work lies in the claim granularity. So we conduct experiments to differentiate granularity first, then evaluate the whole framework.

5.1 Comparing with Other Granularity

Previous works use different granularity for hallucination detection, including response, sentence

[2] https://huggingface.co/ynie/roberta-large-snli_mnli_fever_anli_R1_R2_R3-nli

	Zero Context		Noisy Context		Accurate Context	
	Pearson	Spearman	Pearson	Spearman	Pearson	Spearman
SelfCheckGPT	35.40	43.15	36.31	32.15	40.23	32.55
FActScore	42.58	45.60	33.36	29.91	27.80	27.05
FacTool	59.78	62.57	46.35	38.69	31.41	32.82
REFCHECKER						
Claud 2 + GPT-4	**83.69**	**82.99**	**53.14**	**47.89**	**60.99**	**58.96**
Mistral-SFT + AlignScore	**75.81**	**74.16**	**53.88**	**45.09**	**46.34**	**43.22**

Table 2: Automated checking results of REFCHECKER and previous approaches. The results of REFCHECKER are from the best performing combinations (*extractor* + *checker*) of purely proprietary (blue) and purely open-source models (orange).

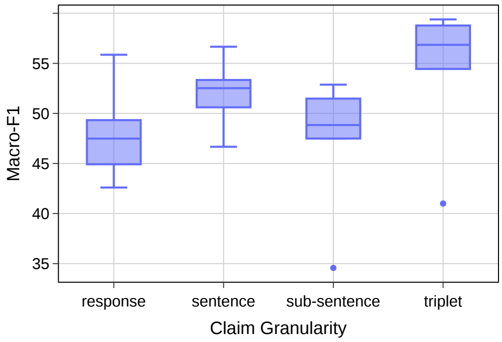

Figure 5: Performance statistics of 7 checkers under different claim granularities on 2.1k manual annotated responses. The detailed checker performance can be found in Table 7 of Appendix B.2.

and sub-sentence levels (cf. Sec. 2 and 3.2). We compare with them to verify the effectiveness of checking on facts with the triplet format.

To make the results with different granularities comparable to each other, we first breakdown the 2.1k annotated responses into different granularities, then collect corresponding checker predictions respectively, and finally aggregate finer-level results all into the response-level. We utilize a strict aggregation rule with zero-tolerance on hallucinations, which means we apply max-pooling (Entailment < Neutral < Contradiction) over claim predictions within a response. We compare the results of 7 checkers, including 4 baseline checkers (RoBERTa-NLI, AlignScore, GPT-4 and Claude 2) and 3 RepC-LE checkers with KNN, SVM and 2-layer MLP classifiers respectively. The evaluation metric is macro-f1 on three categories.

As shown in Figure 5, checking at triplet-level claims is superior over other granularities, with a significant lead against response-level (10 pts macro-f1 score on average). Checking at sentence-level improves over response-level by 5 pts. However, we see a 3.5 pts drop moving to sub-sentence, one of the reasons being sub-sentence claims can overlap. Apparently, the flexibility of sub-sentences leads to poor quality of claim extraction, which subsequently affects checking.

5.2 Comparing with Other Approaches

To contrast with other hallucination detection methodologies, we compare REFCHECKER with three popular approaches on our benchmark, SelfCheckGPT (Manakul et al., 2023), FActScore (Min et al., 2023) and FacTool (Chern et al., 2023). We convert metrics in these approaches to "hallucination rates" as follows:

- SelfCheckGPT: the average score of the sentences within a response. The score is 0, 0.5 and 1 for an `accurate`, `minor_inaccurate`, and `major_inaccurate`, respectively.
- FactScore/FacTool: the proportion of claims not supported (FactScore) or non-factual (FacTool) by the reference.
- REFCHECKER: the proportion of Neutral and Contradiction claims.

Table 2 presents the Pearson and Spearman correlations between the hallucinations rates as evaluated by human and the model results. It reveals that REFCHECKER significantly outperforms previous methods across all three context settings with both proprietary and open-source models. Specifically, the combination of Claude 2 + GPT-4 outperforms the best alternative, FacTool, by 6.8 to 26.1 points. We additionally apply REFCHECKER to SelfCheckGPT's dataset, and observe that in 11 out of the 15 combinations (73%) of REFCHECKER outperform SelfCheckGPT (cf. Table 13 in Appendix C).

5.3 Evaluation on REFCHECKER Framework

To validate the robustness and reliability of REFCHECKER, we checked 7 LLMs' responses on our benchmark and compared rankings (ratios macro-averaged on responses of each LLM) by REFCHECKER and humans with Spearman's rank correlation coefficient. The configuration space consists of different combinations of *extractor* + *checker*, and also the 3 task settings as well as their